



**OMDENA
HYD**

Leveraging LLMs to Understand Global Mental Health Well-being & Fomo in Social Media

MARKET RESEARCH – TEAM 5
Compiled by: Ankitha Beeram

TODAY'S DISCUSSION

TOPICS TO COVER

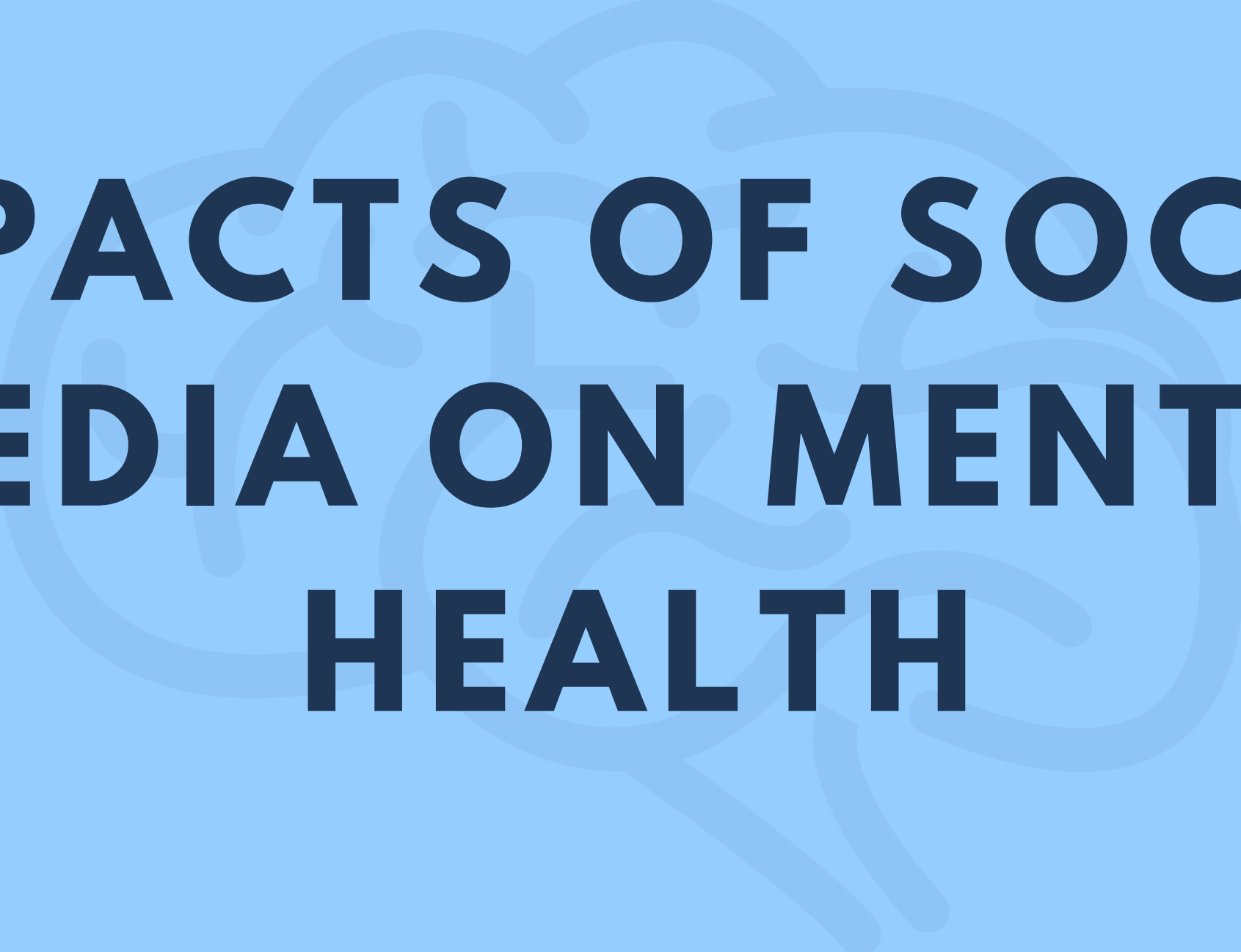
- *Understanding Mental Health – Meriem Baziz*
- *Common Mental Health Disorders – Elizabeth Waithaka*
- *Selected Focus Areas for Project Emphasis – Elizabeth Waithaka*
- *Literature Review – Meriem Baziz*
- *Available Solutions – Victoria Voloshyna*
- *Limitations of current solutions – Vijaya Krishna*
- *Proposed Direction – Krishna Kishore Gajjala*





UNDERSTANDING MENTAL HEALTH

Mental health encompasses a state of well-being where individuals can realize their abilities, cope with life's stresses, work productively, and contribute to their community. Good mental health enables individuals to cope with life's challenges and contribute positively.



IMPACTS OF SOCIAL MEDIA ON MENTAL HEALTH

COMMON MENTAL DISORDERS



1 in 4 people will be affected by mental illness at some point in the lives



13%

The percentage of increase of prevalence of mental health problems per year

300 MILLION

People who suffer from depression, regardless of culture, age, gender, religion, race or economic status.

15-29

AGES

Suicide is the fourth leading cause of death among people aged 15-29 according to Statistics in the US.

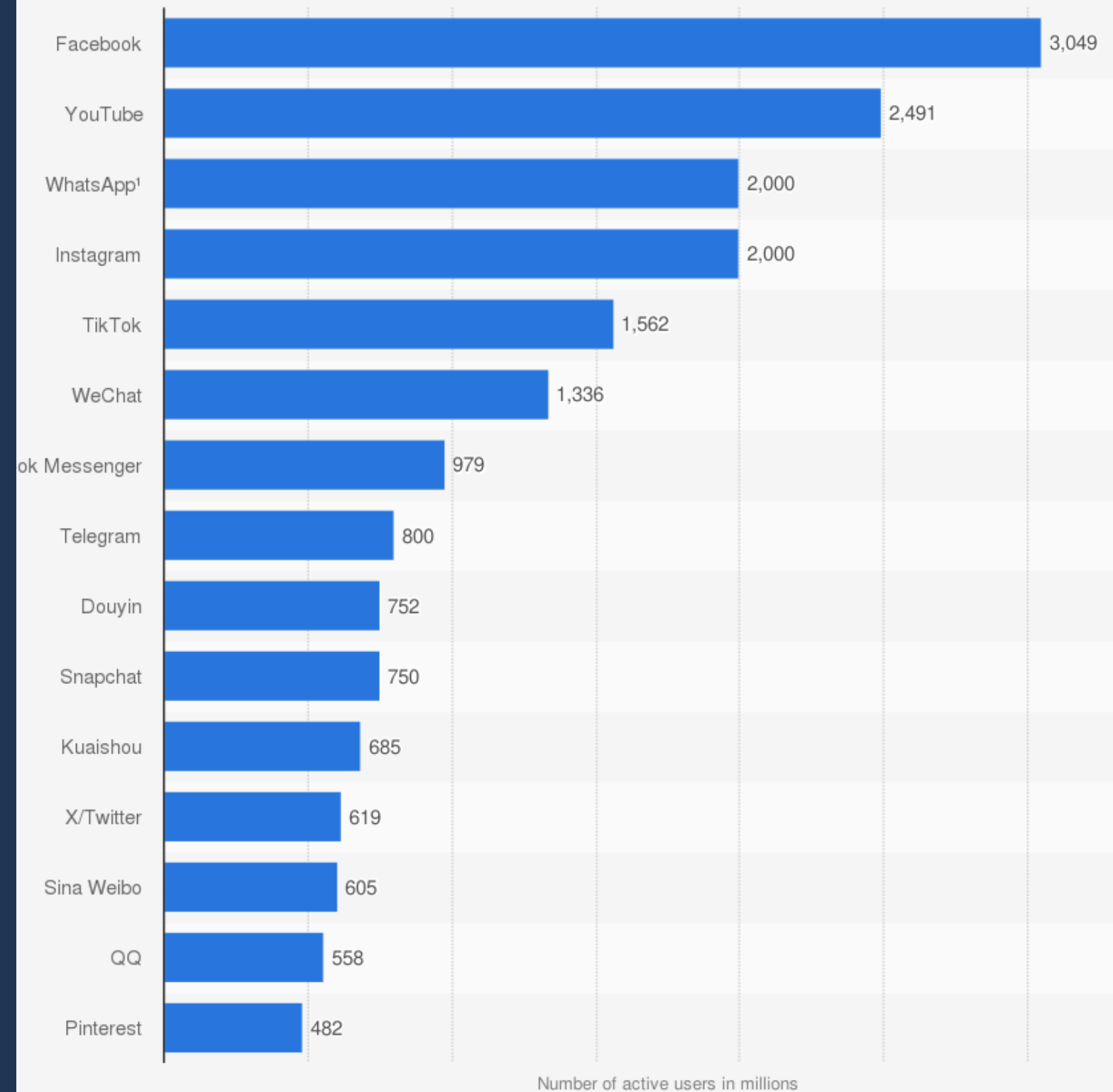
284

MILLION

People who suffer from an anxiety disorder worldwide



Most popular social networks worldwide as of January 2024, ranked by number of monthly active users (in millions)



SOCIAL MEDIA STATISTICS

- Social media platforms enable user interaction through media, text, and calls.
- In January 2024, Kepios reported 5.04 billion global social media users, 62.3% of the total global population.
- Facebook has 3.049 billion active users, followed by YouTube, WhatsApp, and Instagram in worldwide usage.

Mental Health Associated with Social Media Use



- **58%** of Americans report negative impacts on mental health due to social media.
- **64%** of users may experience anxiety disorders from social media,
- **56%** of users experience depression, potentially due to less emotionally satisfying electronic connections, leading to social isolation.
- **60%** of users perceive others' lives as perfect, contributing to self-esteem issues.
- **59%** of users consider cyberbullying a significant concern in social media usage.
- **71%** of users blame misinformation and disinformation for negative mental health impacts.
- **53%** percent more likelihood of experiencing new ADHD symptoms was found among high school students with higher social media use in a University of Southern California study tracking 2,800 participants over two years, all of whom had no previous association with Attention Deficit Hyperactivity Disorder (ADHD).

AREAS OF FOCUS

01

**Dépression
Loneliness**

02

**Attention deficit
hyperactivity
disorder
(ADHD)**

03

**Self
Esteem**

04

**Anxiety
Disorders**

Literature Review

most common use cases
where
LLMs/AI/ML
are helping



■ CONVERSATIONAL AGENTS

[Large Language Models in Mental Health Care: a Scoping Review](#)

■ RESOURCE ENRICHMENT

[Artificial Intelligence and the Mental Health Space: Current Failures and Future Directions](#)

■ WEARABLE DEVICES AND BEHAVIOR MONITORING

[Large Language Models are Few-Shot Health Learners](#)

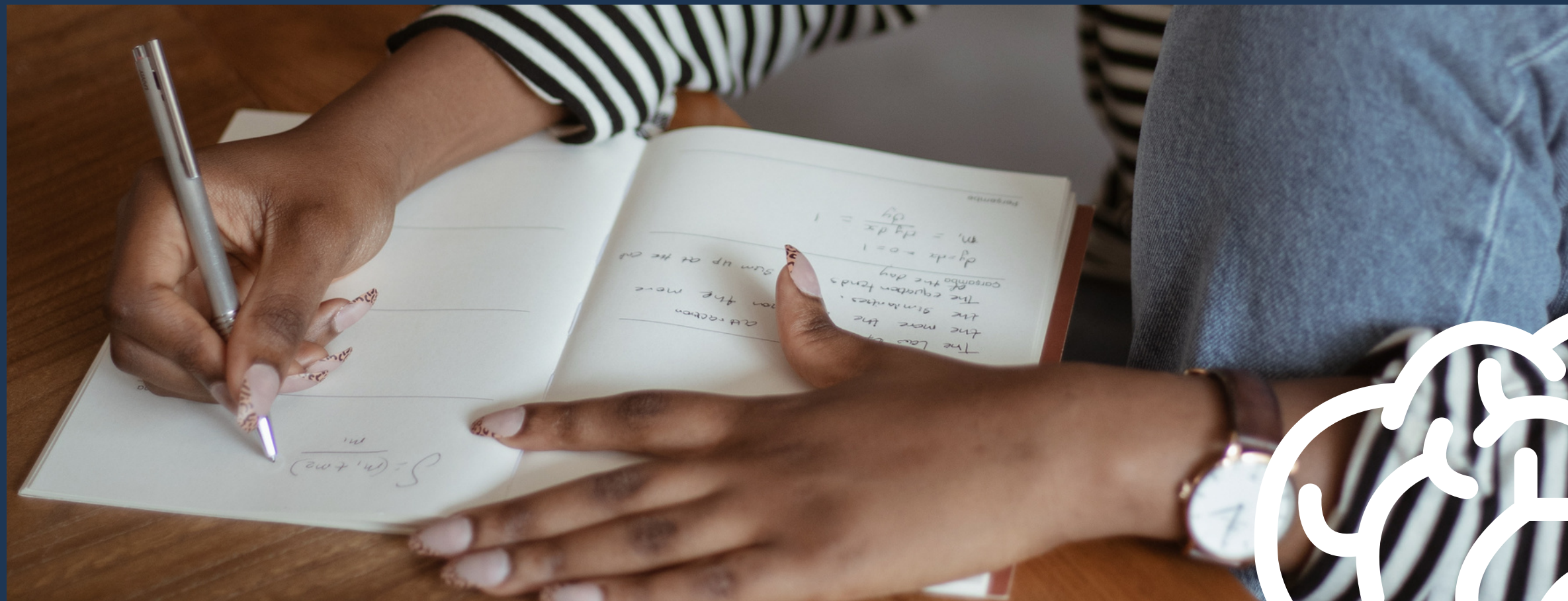
■ SENTIMENT DETECTION AND ANALYSIS

[Read, Diagnose and Chat: Towards Explainable and Interactive LLMs-Augmented Depression Detection in Social Media](#)

■ SIMULATING SOCIAL NETWORKS

[Homophily in An Artificial Social Network of Agents Powered By Large Language Models](#)

SOLUTIONS TO ADDRESS SELECTED MENTAL HEALTH PROBLEMS



- Addressing **Depression**
- Addressing **ADHD**
- Addressing **Self- Esteem Issues**
- Addressing **Anxiety Disorders**
- How LLMs help ?



Solutions to Address **DEPRESSION**



- 1. Limiting Social Media Use**
- 2. Promoting Healthy Habits (including sleep, food)**
- 3. Creative Expression**
- 4. Positive Thinking**
- 5. Seeking Professional Help**
- 6. Cyberbullying Awareness and Prevention**
- 7. Resiliency Building**
- 8. Education and Awareness**
- 9. Support Groups**
- 10. Policy and Regulation**

For more information, check:

[The Social Dilemma: Social Media and Your Mental Health, 5 Ways to Help Yourself Through Depression, SAMHSA's National Helpline, The effects of cyberbullying victimization on depression, Marketing Strategy for Primary Care/Internet-Based Depression Prevention Intervention for Adolescents](#)

Solutions to Address ADHD

1. Prescription medication

(methylphenidate, lisdexamfetamine, dexamfetamine, atomoxetine, and guanfacine)

2. Psychoeducation

3. Behavioral Therapy

4. Additional Lifestyle and Home Remedies

5. Parenting and Family Interventions

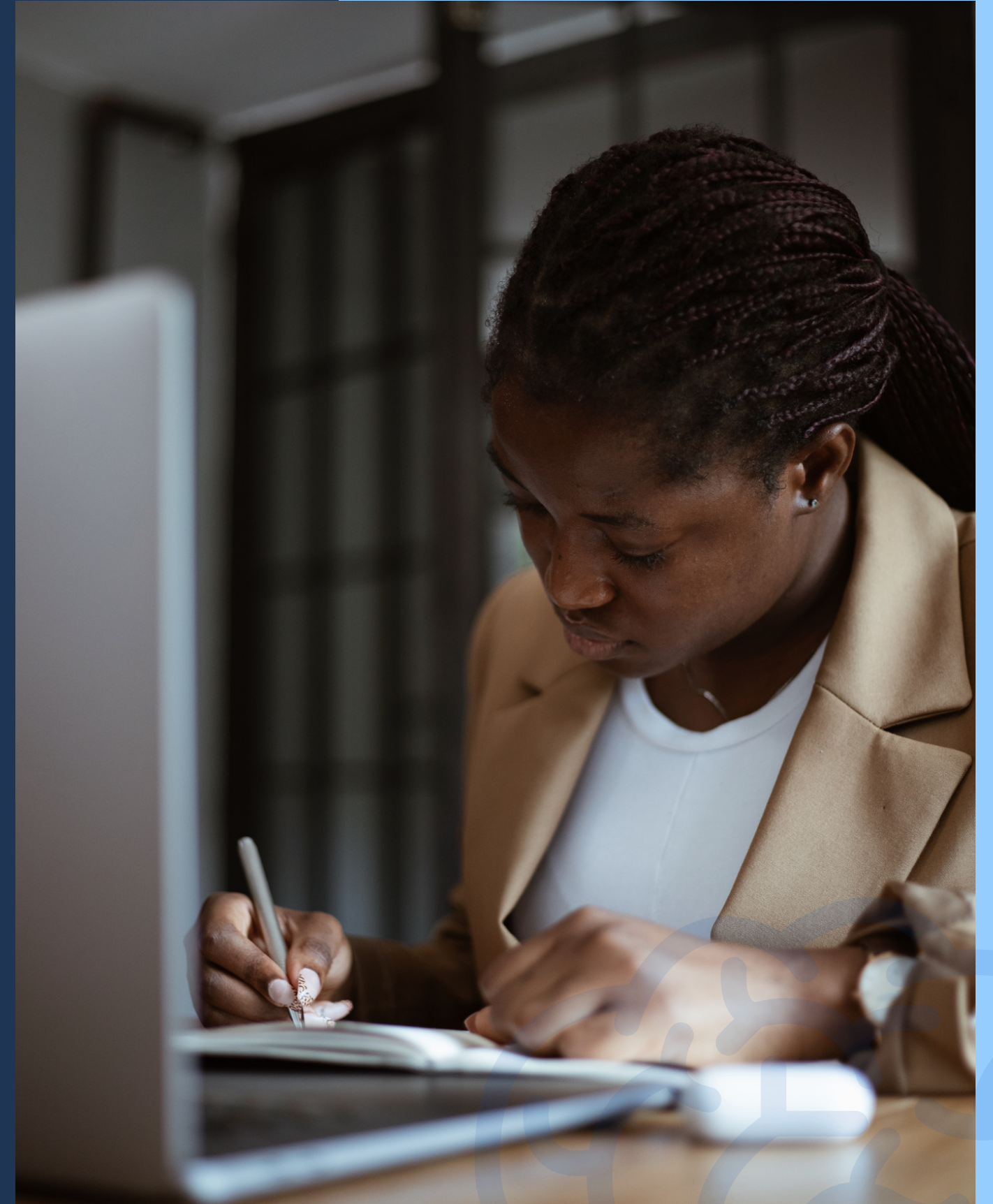
6. Support Groups

7. Peer Interventions

8. Mindfulness Meditation

9. Additional non-prescription Medications

10. Education on Social Media Use



For more information, check: [ADHD](#), [Treatment ADHD](#), [Understanding Problematic Social Media Use in Adolescents with ADHD](#),
[CHADD Affiliates](#)

Solutions to Address SELF-ESTEEM ISSUES

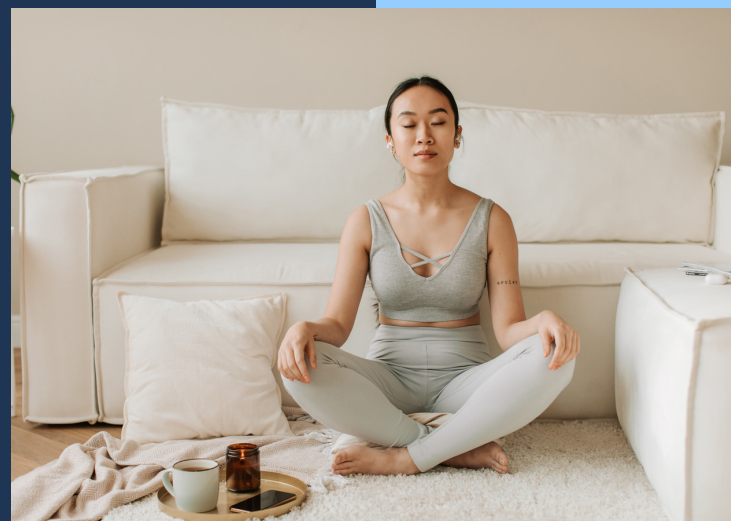


- 1. Self-Reflection**
- 2. Limit Social Media Use**
- 3. Positive Relationships**
- 4. Self-Compassion**
- 5. Assertiveness Training**
- 6. Professional Help**
- 7. Mindfulness and Meditation**
- 8. Hobbies and Interests**
- 9. Education and Awareness**
- 10. Support Groups**

For more information, check: [Self-esteem isn't everything](#), [Social media and low self-esteem](#), [Raising low self-esteem](#), [The relationships between social media use and factors relating to depression](#)

**Solutions
to
Address**

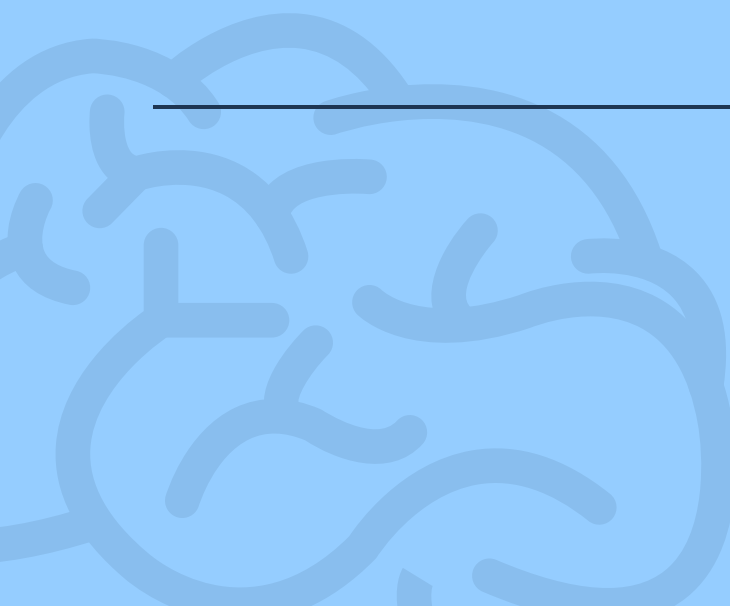
ANXIETY DISORDERS



- 1. Limiting Social Media Use**
- 2. Healthy Social Media Habits**
- 3. Mindfulness and Meditation**
- 4. Professional Help**
- 5. Improve Physical Health**
- 6. Education and Awareness**
- 7. Cyberbullying Prevention**
- 8. Support Groups**
- 9. Improve Communication Skills**
- 10. Upgrade Policies and Regulations**

For more information, check: [The Social Dilemma](#), [The Impact of Social Media Use Interventions on Mental Well-Being: Systematic Review](#), [Relationship between Social Media Use and Social Anxiety in College Students: Mediation Effect of Communication Capacity](#), [Social Media Use and Mental Health: A Global Analysis](#), [StopBullyingGOV](#)

TARGETED INTERVENTIONS AND RECOMMENDATIONS FOR THOSE HAVING MENTAL HEALTH ISSUES CAUSED BY SOCIAL MEDIA



		LLM
1	Incorporating healthy habits (including social media)	✓
2	Building stronger relationships, socialize	✓
3	Getting professional help	✓
4	Meditation, mindfulness	✓
5	Joining a support group	✓

		LLMs
1	Promoting mindful use of social media	✓
2	Reducing cyberbullying through reporting, awareness campaigns	✓
3	Enhancing social media literacy	✓
4	Creating better policies and regulations	✓



Getting healthier online environment

LLM-POWERED APPS FOR FIXING MENTAL HEALTH ISSUES CAUSED BY SOCIAL MEDIA (100+)



YOUPER AI

coaching



WYSA

coaching, connects to professionals



MINDDOC

recording mood in real time, coaching

BREATHHH

mindfulness, healthy habits

SANVELLO

mindfulness, CBT, meditation, support group

REPLIKA

digital friend

KINTSUGI

talking to get insights on mental health

CALM

mindfulness, managing stress, sleep issues

MOODMISSION

dealing with anxiety, connecting to professionals

ROOTD APP

anxiety relief

Limitations for Existing Solutions

**LIMITED
EMOTIONAL
INTELLIGENCE**

**ETHICAL
RESPONSIBILITY AND
ACCOUNTABILITY**

**INABILITY TO ADAPT
TO INDIVIDUAL
NEEDS**

**LACK OF TRAINING
AND EXPERTISE**

**FLEXIBILITY AND
ADAPTABILITY**





Proposed Direction



PHASE 1: UNDERSTANDING THE LANDSCAPE

- Establish mental health metrics related to social media use.
- Identify data requirements for social media usage and mental health.

PHASE 2: DATA COLLECTION AND PREPARATION

- Collect social media data with privacy and ethical considerations.
- Annotate data for mental health conditions.
- Pre-process and clean dataset, removing PII.

PHASE 3: ANALYSIS, MODEL DEVELOPMENT AND TRAINING

- Sentiment analysis and NLP to identify mental health patterns.
- Use statistical methods to determine if there is a significant relationship between social media usage and mental health outcomes.
- Explore cyberbullying with text classification and anomaly detection.
- Train/fine-tune LLMs with RAG for context understanding

PHASE 4: CHATBOT DEVELOPMENT, INTERVENTION STRATEGIES AND TRAINING

- Develop an empathetic chatbot for voice interaction using LLMs.
- Implement mental health evaluation pipeline evaluation and recommendation pipelines.
- Train the chatbot using datasets, and implement specialized prompts and responses in the chatbot
- Refine intervention strategies and address ethical considerations.
- Test the system for accuracy, responsiveness, and user satisfaction.
- Make iterative improvements based on user feedback and observed interactions.

PHASE 5: TESTING AND REFINEMENT

- User testing to gather feedback and improve chatbot.
- Fine-tune the chatbot's ability to detect signs of mental health issues, cyberbullying and online harassment.

PHASE 6: EVALUATION AND DEPLOYMENT

- Evaluate chatbot performance using established metrics.
- Deploy and monitor the chatbot, ensuring ethical, privacy and security.
- Adjust based on real-world efficacy.





FINAL DELIVERABLES

- Dataset annotated with mental health indications.
- LLM-based chatbot/mobile app for voice interaction.
- Mental health evaluation and recommendation pipeline.

PARTICIPANT LEARNING OUTCOMES

- Practical experience in AI areas: LLMs, RAG, Prompt Engineering, correlation analysis, sentiment analysis, semantic search, chatbot development, ethical and responsible AI.

TEAM 5

PROJECT MANGER:

Krishna Kishore Gajjala
(Co-PM: Ankitha Beeram)

DATA-COLLECTION:

Lead: Elizabeth Waithaka

DATA-PREPROCESSING

Lead: Victoria Voloshyna, Co-Lead: Sayak Mallick

MODEL-BUILDING:

Lead: Debanjan Mondal, Co-Lead: Frankline Ononiwu



**THANK
YOU**